CS248 : 01_ Probability and Linear Algebra

(Review) :

- Random variable : X (set of possible values)

↳ Probability dist. with Random variable X :

$$P(X = x) = P(x) = P_X(x)$$

discrete r.v. : Probability mass function : (pmf)

$$\sum_{x \in X} P(X=x) = 1$$

continuous r.v. : Probability density function

$$\begin{cases} P_X(x) \text{ is the } \underline{\text{likelihood}} \text{ of } x \\ P_X(a \le x \le b) = \int_a^b f(x)\,dx \\ \int_{\alpha} f(x)\,dx = 1 \end{cases}$$

- Bernoulli dist. $\{x \mid 0, 1\}$

$P(x=1) = \pi \quad P(x=0) = 1-\pi$

$P(x) = \pi^x (1-\pi)^{1-x}$

Poisson dist. # of arrivals over fixed time period. $\mathcal{X} = \{0, 1, 2, \cdots\}$

λ : average arrival rate $\lambda \ge 0$

$$P(x) = \frac{1}{x!} e^{-\lambda} \cdot \lambda^x$$

(x factorial)

Normal dist (Gaussian) :

$X = \mathbb{R} \quad (\mu, \sigma^2)$

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

joint dist. $P(X=x, Y=y) = P(x, y)$

⇓ Marginal dist

$$P(X=x) = \sum_y P(X=x, Y=y)$$

$$P(x) = \int_{y \in Y} p(x, y)\,dy$$

Conditional Probability :

$$P(x \mid Y) = \frac{P(X=x, Y=y)}{P(Y=y)}$$

↓ condition

$X \perp\!\!\!\perp Y$ iff $p(x \mid y) = p(x)$

⇓

$p(x, y) = P(x) P(y)$

Bayes Rule :

$$P(y \mid x) = \frac{P(x \mid y)\, p(y)}{\int p(x \mid y)\, p(y)\, dy} \sim p(x)$$

Expectation :

$E[X]$:

$$E_X[f(x)] = \sum f(x) p(x)$$

$$E_X[f(x)] = \int f(x) p(x)\, dx$$

$$E[X] = \int x p(x)\, dx$$

$$Var(X) = E[(X-\mu_x)^2] = E[X^2] - E[X]^2 = \sigma_x^2$$

$$Cov(X, Y) = E[(X-\mu_x)(Y-\mu_y)]$$

$$= \iint_{xy} (x-\mu_x)(y-\mu_y)\, p(x, y)\, dx\, dy$$

$$= E[XY] - \mu_x \mu_y = \sigma_{xy}^2$$

- Vector :

$$\begin{cases} x = [x_1 \cdots x_d]^T \\ A = [a_1 \cdots a_n] = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \end{cases}$$

$$x^T y = \sum_i x_i y_i \qquad \|x\| = \sqrt{x^T x} = \sqrt{\sum_i x_i^2}$$

⇓

$$d(x, y) = \|x - y\| = \sqrt{\sum (x_i - y_i)^2}$$

$$x y^T = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} [y_1 \cdots y_d] = \begin{bmatrix} x_1 y_1 & \cdots & x_1 y_d \\ \vdots & & \vdots \\ x_d y_1 & \cdots & x_d y_d \end{bmatrix}$$

$x \in \mathbb{R}^n$.

$$\begin{cases} y = Ax = \sum_i \underset{\text{vector}}{a_i} x_i \\ y = A^T x = \begin{bmatrix} a_1^T \\ \vdots \\ a_n^T \end{bmatrix} [x] = \begin{bmatrix} a_1^T x \\ \vdots \\ a_n^T x \end{bmatrix} \end{cases}$$

Vector : $x = (x_1 \cdots x_n)^T$

$p(x) = P(x_1 \cdots x_n)$ is a joint dist.

$$\mu = E[x] = \int x p(x)\, dx = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix}$$

$$Cov(X) = E[(X-\mu)(X-\mu)^T] = E[XX^T] - \mu\mu^T$$

⇓

a matrix

multivariate Gaussian:

$x \in \mathbb{R}^d$. $\mu \in \mathbb{R}^d$

cov matrix $\Sigma \in S_{++}^d$.

$$p(x) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}\|x-\mu\|_\Sigma^2}$$

$$= N(x|\mu, \Sigma)$$

$$\|x-\mu\|_\Sigma^2 = (x-\mu)^T \Sigma^{-1} (x-\mu)$$

$|\Sigma|$ = volume of Gaussian.

- special: Diagonal $\Sigma = \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_2^2 \end{bmatrix}$

$$N(x|\mu, \Sigma) = \prod_{i=1}^{d} N(x_i|\mu_i, \sigma_i^2) \quad |\Sigma| = \prod \sigma_i^2$$

d independent Gaussian.

- $y = ax + b$

  $E[y] = aE[x] + b$

  $var(y) = a^2 E[x]$

  $y = Ax + b$

  $E[y] = AE[x] + b$

  $cov(y) = A\,cov(x)A^T$

if $x+y$ ($p(x,y) = p(x)p(y)$) Independent

$E[xy] = E[x]E[y]$

$cov(x,y) = 0$

Inpendent $\Rightarrow$ uncorrelated.

- $N(x|a, A)\,N(x|b, B) = Z\,N(x|c, C)$

$$c = C(A^{-1}a + B^{-1}b)$$

$$C = (A^{-1} + B^{-1})^{-1}$$

$$Z = \frac{1}{(2\pi)^{d/2}|A+B|^{\frac{1}{2}}} e^{-\frac{1}{2}(a-b)^T(A+B)^{-1}(a-b)}$$

$$= N(a|b, A+B)$$

$$\downarrow \int N(x|a, A)\,N(x|b, B)\,dx = N(a|b, A+B)$$

- $f(x) = x^T A x - 2x^T b + c$

  $= (x-d)^T A (x-d) + e$

  $d = A^{-1} b$

  $e = c - d^T A d = c - b^T A^{-1} b$.

- λ_i are the eigenvalues of $A^{n\times n}$

  $tr(A) = \sum \lambda_i$

  $|A| = \prod \lambda_i$

  $\{\lambda_i, x_i\}$ are also eigenvalue/vector of inverse A^{-1}

$\Downarrow$

positive definite:

$\{\lambda_i, x_i\}$ of A: $\lambda_i > 0$

for all matrix $A^{m\times n}$

$G = A^T A$ is positive definite.